Tradable services and local job multipliers: evidence from Brazil

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August 19, 2026

Abstract

The rapid growth of cross-border trade in services, driven in large part by digital technologies that have lowered the cost of supplying services across distance, has transformed the nature of globalisation. Despite this transformation, little is known about the local labour market effects of employment growth in tradable service activities. This paper sheds new light on how expansions in tradable services employment spill over onto jobs in non-tradable services. Drawing on matched employer–employee administrative microdata covering the universe of formal workers in Brazil between 1995 and 2023, we construct a granular classification of tradable and non-tradable services based on geographic concentration. We then estimate local job multipliers using a shift-share instrumental variable design applied to an annual panel of local labour markets, exploiting growth in tradable services subsectors. Our results indicate that tradable services generate substantial local spillovers: for every additional job in tradable services, approximately 1.3 jobs are created in non-tradable services. This effect is larger than that of the overall tradable sector. These spillovers operate primarily through household demand, being concentrated in activities catering to consumers rather than to firms. While our empirical design does not directly observe trade flows, the findings are consistent with the growing importance of service activities that can serve non-local demand in shaping local labour market dynamics. These results highlight the need to incorporate tradable services into discussions of regional development and employment creation.

Keywords: Job multipliers, local labour markets, tradable services

JEL classification: F16, R12, R23

1 Introduction

Over the past two decades, global trade in services has grown rapidly, supported by the diffusion of digital technologies that have reduced the cost of delivering services across distance (Benz et al., 2022; Herman and Oliver, 2023; WTO, 2019). These developments have expanded the set of services that can be supplied beyond local markets, blurring the traditional distinction between tradable and non-tradable activities. Despite this transformation, much of the academic and policy debate on the labour market effects of trade continues to focus on manufacturing and to treat services as predominantly local, non-tradable activities.

This characterization is increasingly at odds with both theory and evidence. A growing share of services — such as finance, architectural services, and activities related to computer programming — can now serve non-local demand, whether across regions within a country or across borders. Several studies document that services have become more spatially concentrated within countries over the past few decades (Chen et al., 2023; Desmet and Rossi-Hansberg, 2014), which is partly related to a reduction in the so-called "proximity burden": the need for co-location of sellers and buyers of services (Francois, 1990). Rising tradability and rising concentration together imply that jobs in certain services sectors can now be created through external demand, raising an important question about the local labour market consequences of such job growth.

This paper studies the local employment spillovers generated by employment growth in tradable services sectors. In particular, it asks whether expansions in these sectors — which are capable of serving non-local demand — generate additional employment in local non-tradable activities. The quantification of such spillovers is commonly referred to as the job multiplier, defined as the number of additional jobs created in the non-tradable sector for each new job in the tradable sector in a local labour market.[1] The modern empirical literature on job multipliers builds on the seminal study by Moretti (2010), who uses a Bartik (1991) shift-share instrument to measure how many additional jobs in non-tradable sectors were created for every new job in manufacturing in US metropolitan statistical areas. Since then, most studies have continued to focus on manufacturing job multiplier effects, providing evidence that new employment is created as a result of job growth in the tradable sector in the United States (Gerolimetto and Magrini, 2016; Nguyen and Soh, 2017), Europe (Cerqua and Pellegrini, 2020; de Blasio and Menon, 2011; Goos et al., 2018a), China

[1] Multiple terms can be used to refer to local labour markets, which can vary by country of analysis. For the purpose of this study, local labour markets will also be referred to as regions or, empirically, as microregions.

(Wang and Chanda, 2018) and Japan (Kazekami, 2017). Recent studies have acknowledged the importance of tradable services and have integrated them into a broader traded sector alongside manufacturing (Frocrain and Giraud, 2019; Osman and Kemeny, 2022; Van Dijk, 2018). In a rare exception, Avdiu et al. (2022) examine the job multiplier effects of tradable services alone, in the case of India.

Three gaps follow from this. First, there is still little evidence on the role of tradable services exclusively as a driver of job growth in the non-tradable sector. Second, the evidence base remains concentrated in advanced economies, which may not generalise to the structure of local labour markets in emerging economies, where structural characteristics such as spending patterns, informality and employment protection differ. Third, the understanding of mechanisms underlying job multiplier effects remain limited, with existing evidence on mechanisms largely focused on spillovers from high-technology manufacturing (Goos et al., 2018b; Lee and Clarke, 2019; Moretti, 2010).

As a large emerging economy with specialised and economically distinct local labour markets, Brazil has also attracted attention in the literature. Four studies apply the job multiplier framework to Brazilian regions, examining manufacturing (Macedo and Monasterio, 2016; Rocha and Araújo, 2021), public sector employment (Loyo et al., 2018), and sectoral complexity (Queiroz et al., 2026), but none has looked at tradable services. These studies also share a common design, relying on two or three time points in long differences. The multipliers they report, ranging from roughly 1.8 to above 6, are substantially larger than those typically reported for advanced economies.

Yet Brazil also provides a particularly relevant setting for studying the services sector. Services have been an important source of employment growth: formal employment in the tradable services sectors grew by 177% between 1995 and 2023, compared with 153% in non-tradable services and 133% in the economy as a whole.[2] Brazil is also unusually service-oriented for a large emerging economy, with a services sector that accounts for over 70% of total employment, compared with 46% in China, 30% in India and 62% in Mexico. Its employment structure is therefore closer to that of advanced economies such as Germany (72%) and the United States (79%).[3] The size and dynamism of the Brazilian service economy therefore make the question of whether tradable service employment generates broader local employment particularly relevant.

Focusing on Brazil, this study combines rich administrative data with recent methodological

[2]Figures are calculated from the Brazilian RAIS dataset.

[3]Figures refer to 2022 and are calculated from ILOSTAT (Employment by sex and economic activity, International Labour Organization).

advances to analyse the job multiplier associated with tradable services. Using matched employer–employee data covering the universe of Brazilian formal employment, we construct a granular classification of tradable and non-tradable services and exploit variation in employment growth across local labour markets. We build on the established job multiplier framework while using a panel shift-share instrumental variable design to isolate variation in tradable services employment driven by external demand. This approach allows us to move beyond the long-difference designs that characterize much of the existing literature and to exploit multiple periods of data while accounting for time-varying shocks and local labour market heterogeneity (Bartik and Sotherland, 2019; Borusyak et al., 2025; Osman and Kemeny, 2022). The paper therefore provides evidence on job multipliers generated specifically by tradable services in an emerging economy, while also employing a panel-based shift-share approach in the job multiplier literature.

We find that, for every new job in tradable services, 1.3 additional jobs in non-tradable services are created in Brazil, at least in the short term. This multiplier is larger than that estimated for the tradable sector as a whole and comparable to estimates reported in the existing literature, while smaller than previous estimates for Brazil. These effects tend to be stronger for men and younger workers, and the evidence points to increased household demand as the main mechanism underlying the employment response. We also find that the results are not driven solely by large cities such as São Paulo. Finally, spillovers are not confined to employment: hours worked and wages of workers in non-tradable services also increase alongside job growth in tradable services.

The remaining part of this paper is organized as follows. Section 2 reviews theoretical concepts and the current empirical literature on trade and labour markets. Section 3 outlines data sources and estimations, as well as descriptive statistics on tradable and non-tradable services and the final sample. Section 4 describes the empirical strategy, and Section 5 presents results. Section 6 concludes.

2 Conceptual framework

Job multiplier analysis is inserted in the broader literature of the impact of trade on labour markets. Most studies in this field focus on direct effects of trade exposure on workers in the exposed sector or region. The seminal constribution of Autor et al. (2013) paper illustrates a common research question, namely estimating the impact of import competition — the China shock —

on manufacturing workers. Studies in the field broadly agree on the persistent employment and wage losses in exposed sectors and regions (Dix-Carneiro and Kovak, 2017; Malgouyres, 2017). More recently, important nuances to such movements have been made to account for the significant portions of workers that reallocated from manufacturing to services in developed economies (Bloom et al., 2024). On a different angle of trade exposure, increased export opportunities are found to translate into increased worker's wage and within-sector firm switching (Dauth et al., 2021).

Job multiplier analysis differs from these studies by focusing on spillover effects on workers not directly exposed to trade. Theoretically, job multipliers should be interpreted within a spatial general equilibrium framework (Moretti, 2010). In this setting, the economy consists of multiple regions (equivalent to local labour markets) producing two types of goods and services: tradables, whose prices are determined in external markets, and non-tradables, whose prices and quantities are determined locally. Regions are populated by mobile workers who choose locations based on wages, prices, and idiosyncratic location preferences, implying an elastic local labour supply. Housing supply is also elastic but constrained by local land-use regulations and geographic characteristics, which affect the extent to which increases in labour demand translate into population growth rather than higher local prices.

An exogenous increase in demand for a region's tradable output — such as a rise in foreign demand for a tradable service — raises labour demand in the tradable sector. This shock leads to a direct increase in local employment in a given region, which may be met through migration to the region, increased labour force participation, or reallocation across sectors within the region. To the extent that the employment expansion is not fully offset by sectoral reallocation, net local employment and regional household income increase.

Beyond this immediate rise in local employment in the tradable sector, there are indirect spillover effects in the non-tradable sector, commonly referred to as job multiplier effects. A first mechanism operates through the income or consumption channel: as employment and wages in the tradable sector rise, higher household income increases demand for locally produced services, such as personal care, retail, and hospitality. A second mechanism reflects local input-output linkages, whereby expanding tradable firms increase demand for locally supplied intermediate services, such as security or maintenance. A third channel operates through agglomeration and productivity effects, as increased spatial concentration can raise productivity and employment across sectors (Beaudry and Schiffauerova, 2009).

As highlighted by Moretti and Thulin (2013), the magnitude of the multiplier depends on the strength of these channels. For instance, the extent to which higher household income translates into additional non-tradable employment depends on wage levels in the tradable sector and on local consumption patterns. At the same time, general equilibrium adjustments can attenuate multiplier effects. A constrained housing supply may lead to rising living costs, limiting in-migration and pricing low-wage workers out of the region. Similarly, tight labour markets — due to low unemployment or limited worker mobility — may cause increases in demand for the non-tradable sector to be absorbed through higher wages rather than additional employment.

A further set of forces that may limit the effect of the job multiplier operates on the geography of consumption itself rather than on the size of the income shock. The consumption channel rests on the premise that a rise in local household income is spent locally, yet the share of expenditure that remains within the local labour market is not fixed. The diffusion of e-commerce allows households to direct a growing portion of their spending towards sellers located elsewhere, weakening the link between local income and local retail employment even when the underlying income shock is unchanged. Teleworking operates on the same margin in a different direction. Where the workers producing tradable services no longer commute to a central business district, their consumption of food, personal services and retail shifts towards their place of residence, redistributing non-tradable employment across local labour markets rather than reducing it in aggregate. Althoff et al. (2022) document precisely this mechanism in the United States, showing that the withdrawal of high-skill service workers' local spending during the pandemic fell heavily on consumer service employment in large cities. To the extent that the technologies raising the tradability of services are also those reshaping where the workers producing them spend their income, the estimated multiplier reflects the net of these movements.

Many studies have tried to quantify the job multiplier in different settings, oftentimes only taking the manufacturing sector as the tradable sector. Moretti (2010) initially found a manufacturing job multiplier of 1.6 in US metropolitan areas, rising to 4.9 for high-technology manufacturing. Since, methodological improvements have been added to the framework, pointing to a potential overestimation from this first exercise. For example, excluding the metropolitan area under observation in the calculation of the SSIV (leave-one-out approach); and using metropolitan areas where people work, rather than live. More recently, analysing US commuting zones, Bartik and Sotherland (2019) arrive at a manufacturing multiplier of 0.1-0.8, and a high-technology manufacturing multiplier of 0.9-1.0. Closer to our analysis, Van Dijk (2018) estimates a job multiplier of the whole US tradable sector (tradable services and manufacturing) between 0.17-0.88. Frocrain and Giraud (2019) estimate a similar figure of 0.64 for the whole of the French tradable sector. Methodologically closer to our approach, Osman and Kemeny (2022) report a job multiplier of 0.86 in the US tradable sector. Finally, Avdiu et al. (2022), specifically analysing tradable services in India, find that a 10% increase in tradable services employment leads to a 4.2% increase in non-tradable services employment.[4]

3 Data

Our primary data source is the Relação Anual de Informações Sociais (RAIS), an administrative dataset collected annually since 1986 by the Brazilian Ministry of Labour. RAIS functions as a census of the Brazilian formal labour market, as all registered firms are legally required to report their workforce in order for employees to access social security, severance pay, and other benefits. Non-compliance can result in fines, which creates strong incentives for accurate reporting by both firms and workers. This institutional framework ensures the reliability of the database and explains why RAIS has become the main source for labour market studies in Brazil (Dix-Carneiro and Kovak, 2017; Menezes-Filho and Muendler, 2011).

RAIS covers all formal employees in Brazil, i.e. workers with a signed work card (*carteira de trabalho assinada*), which guarantees access to the benefits of the country's labour legislation. The dataset excludes informal workers, self-employed individuals, domestic workers (with some exceptions), interns, and other minor employment categories. Each observation corresponds to an employer–employee relationship in a given year, and a single worker may appear multiple times if employed by different firms. We measure employment using only relationships active on December 31, which yields an annual stock of formal employment consistent with official published figures and avoids double counting. While RAIS was established in Brazil in 1975, the information it gathers has increased in breadth across time. In particular, industry information became available beyond a coarse 25-sector breakdown only after 1994, which is why we work with RAIS from 1994 through 2023 at the ISIC Rev. 4 2-digit level, totalling 88 unique sectors, 51 of them in services.[5]

[4] In their study, only elasticities are reported. The estimation also differs methodologically from most of the literature, as they approach the shift aspect of the shift-share instrumental variable through services trade data, rather than employment data for tradable services.

[5] Industry information in RAIS is reported under the Brazilian national classification of economic activities (CNAE), which we map to ISIC Rev. 4 divisions using IBGE's official correspondence tables. At the division

Because of its administrative nature, RAIS also allows for the observation of employment at fine geographic levels, for over 5,500 municipalities, which we aggregate to 510 microregions (local labour markets) with stable boundaries over time using IBGE's official crosswalk. Brazilian microregions have been typically used in the literature as proxies for local labour markets (Costa et al., 2016; Dix-Carneiro and Kovak, 2017; Kovak, 2013), since they provide a balance between geographic granularity and the size required to capture the economically contiguous concept of local labour markets. In addition, RAIS provides rich sociodemographic information on workers, enabling heterogeneity analyses beyond aggregate employment. At the same time, it is representative at all levels, since it accounts for the whole population of formal employees. Crucially, RAIS enables the observation of employment at a high-frequency, across all years of the dataset.[6]

The most significant limitation of RAIS is its exclusion of informal workers, who represent a substantial share of total employment in Brazil. Estimates suggest that between 40% and 50% of workers were informally employed during the 2010s, depending on the data source (ILOSTAT, IPEA). Informality is especially prevalent in agriculture, construction, and personal services, while manufacturing and public sector employment are relatively formalized. For example, in 2019, informality rates exceeded 75% in agriculture and over 50% in several non-tradable service industries, but were close to 8% in manufacturing overall.

These patterns are important when interpreting results, as our estimates capture multipliers in the formal sector, and therefore could be reflecting workers moving from the informal sector into the formal sector, rather than new jobs being created. The same limitation operates in the opposite direction as well: because RAIS records only formal contracts, employment created informally in response to a tradable services expansion is invisible to our estimates. This concern is not confined to unregistered firms: as Ulyssea (2018) shows, informality in Brazil operates along two margins, and registered firms routinely hire workers off the books alongside their formal payroll. A firm that reports to RAIS can therefore absorb a local demand shock without any of that adjustment appearing in the data. Informality is moreover concentrated precisely in the activities where the multiplier is expected to operate, since personal services, construction and small-scale retail respond quickly to local demand and do so in large part through informal arrangements. Formalisation and

(2-digit) level, CNAE 2.0 is equivalent to ISIC Rev. 4; for CNAE 1.0, reported in RAIS from 1994 to 2006, we make use of the crosswalk to CNAE 2.0 from IBGE.

[6] As discussed, many studies in the job multiplier literature rely on cross-sectional data, oftentimes census data, that is either infrequent by design, or not robust within short time spans. For example, in France or the United States, census data is collected in a rolling basis, and yearly data also groups information from previous and following years. Which is why comparing employment trends from one year to the other can be highly misleading.

unrecorded informal hiring thus bias the estimated multiplier in opposite directions relative to its total employment counterpart. To control for this, in Section 5.4 we make use of the Brazilian decennial Census (IBGE) which includes both formal and informal workers. While it is infrequent and less detailed by nature, it allows for a holistic view of the Brazilian labour market.

3.1 Defining tradable services

To measure job multiplier effects, we categorise 2-digit service industries into tradable and non-tradable services. In the context of this study, tradable services are services that can meet non-local demand, i.e. beyond the demand of their local labour market. For example, an architect can email plans to a client abroad, while a hairdresser cannot cut a client's hair from afar. This implies that the analysis defines tradable services as those that are mostly deliverable remotely, corresponding to Mode 1 in the WTO context[7] – while also considering within-country trade flows.

Technological progress has lowered the cost of delivering services remotely, increasing their tradability and enabling not only cross-border transactions but also domestic exchanges across regions of the same country. The classification used here therefore rests on whether an activity is capable of serving demand originating outside its own local labour market, irrespective of whether that demand is domestic or foreign. A service delivered from São Paulo to Rio de Janeiro and one delivered from São Paulo to Lisbon are treated alike as tradable, and the spillovers we estimate should accordingly be read as arising from exposure to non-local demand broadly rather than from international trade specifically. Banking illustrates how this capacity has shifted within countries. Serving a client once required a branch in every market, whereas account management, credit assessment and customer support are now routinely supplied from a small number of centralised locations to clients throughout the national territory. Comparable shifts have taken place in accounting, insurance claims processing and technical support, activities of limited tradability two decades ago that are now regularly delivered at distance.

Within this concept, the classification of tradable services in this study builds on Jensen and Kletzer (2005),[8] and measures the geographic concentration of a given sector's (ISIC Rev. 4 2-digit)

[7]The WTO categorises trade in services within four modes of supply: Mode 1, where supplier and consumer remain in their respective territories and similar to traditional trade in goods; Mode 2, where the consumer goes to the supplier, commonly reflecting tourist expenditures; Mode 3, where the supplier goes to the consumer through commercial presence, reflecting foreign direct investment and foreign affiliates; and Mode 4, where the supplier goes to the consumer through the presence of natural persons.

[8]The exercise of identifying which services are tradable has been extensively discussed in the literature, that generally derives tradability from how intensely service sectors are traded (Avdiu et al., 2022; Calligaris et al., 2024);

employment in a given microregion through the following formula:

$$geoconc_i = 100 * \sum_c (S_{ci} - S_c)^2 \tag{1}$$

Where S_{ci} is the share of microregion c in national employment of industry i, and S_c is the share of microregion c in total national employment. The index rises when an industry's employment is concentration in a few microregions beyond what their overall economic size would imply, and industries with high levels of $geoconc_i$ are considerered as geographically concentrated.

Following Frocrain and Giraud (2019), we define the cutoff for tradability as the lowest geographic concentration in a manufacturing 2-digit sector. The cutoff is computed for every year of the dataset, and sectors whose classification is not constant over time are assigned manually on the basis of their ISIC Rev. 4 sectoral descriptions.

Under this classification, tradable services include the following ISIC Rev. 4 sections: information and communication, finance and insurance, real estate fully; and transportation and storage, science and technology and administrative services partially. Non-tradable services fully include construction; wholesale and retail; accommodation and food; public administration, education, and health; arts, and entertainment; and other service activities.[9]

3.2 Descriptive statistics

Table 1 showcases differences between tradable and non-tradable services in the Brazilian formal labour market. Firstly, there are significantly more Brazilian employees in non-tradable services (NTS), at approximately 35 million compared to 10 million in tradable services (TS). However, this gap has decreased over time, since growth in TS has outpaced that of NTS, especially since 2010. The reasons for a significantly higher relative growth rate in TS since 2010 are twofold: firstly, it reflects the rise in tradability of these services, and therefore of services trade; additionally, it excludes the early 2000s, during which took place a large movement of workers - particularly in

the geographical concentration of production, and sometimes consumption, of services (Frocrain and Giraud, 2019; Gervais and Jensen, 2019; Jensen and Kletzer, 2005; Van Dijk, 2018); or from a conceptual analysis based on sectoral descriptions (WTO, 2023). Our approach falls within the second of these, using the spatial distribution of employment to infer whether an activity can serve demand beyond its own market. Classifications nonetheless differ across studies applying similar logic, reflecting both the choice of concentration measure and genuine cross-country differences in the geography of service provision.

[9]A full breakdown of 2-digit ISIC Rev.4 sectors by geographic concentration and category is presented in Table A.1.

NTS - from the informal to the formal sector (Berg, 2010).[10] The exclusion of informal workers also likely leads to an overestimation of the percentage of workers with a college degree, particularly in NTS. In terms of other workforce characteristics, TS tends to employ younger workers and is more male-intensive than NTS. Crucially, as expected from our conceptual and empirical approach to categorising services, TS tend to be significantly more concentrated in the Brazilian territory.

Table 1: Tradable and non-tradable services in the Brazilian formal labour market

Variable	TS	NTS
Million employees in 2023	9.6	34.7
Employment growth between 1995 and 2023	177%	153%
Employment growth between 2010 and 2023	54%	23%
% of over 45-year-olds	26%	34%
% of female workers	41%	50%
% of workers with college degree	25%	25%
HHI of employment across microregions ($\times$ 100)	74	30

Note: The HHI (Herfindahl–Hirschman index) measures concentration of employment shares across microregions, scaled by 100. This differs from the classification index in equation 1, which measures the deviation of an industry's regional employment distribution from the distribution of total employment.

Table 2 reports the final sample in the analysis, which is composed of a panel of 510 microregions across 29 years (1995-2023).[11] As expected, microregions differ significantly in terms of size. With a population largely concentrated in coastal areas, Brazil's microregions have a median employment of 63 thousand workers, with a mean of nearly 200 thousand workers. Large local labour markets, such as those centered around São Paulo or Rio de Janeiro, significantly skew the distribution of employment in Brazil. This is particularly true for tradable services employment.

The average yearly growth rate of TS employment stood at 6%, outpacing both NTS and total employment growth. This gap is consistent with the aggregate patterns reported in Table 1, and it is the differential expansion of tradable services across microregions that the empirical strategy set out below seeks to exploit.

[10] While this change has been associated with the Brazilian economic growth during the 2000s commodity boom, it is also a results of policy. For instance, Abras et al. (2018) show that, in early 2000s Brazil, increased enforcement of labour market regulations in cities led to an increase in gross and net formal job creation.

[11] As is described in Section 4, the shift-share instrument approach requires excluding the first year from the final sample.

Table 2: Sample statistics (510 microregions, all years pooled)

Variable	Mean	Median	SD
Total employment (incl. agriculture and mining), thousands	195.8	63.1	725.5
Tradable employment, thousands	38.2	12.8	158.6
Tradable services (TS) employment, thousands	18.8	3	113.5
Non-tradable services (NTS) employment, thousands	69.1	20.9	262.9
Avg yearly employment growth rate	5.07	4.45	2.85
Avg yearly NTS employment growth rate	5.48	4.70	2.98
Avg yearly TS employment growth rate	6.11	5.90	2.67
Avg NTS real hourly wage	13.31	13.20	2.12
Percentage of informal workers (Census 1991)	73.6	76.8	15.7
Percentage of informal workers (Census 2010)	62.0	60.5	14.9
Avg yearly TS employment growth rate (SSIV)	3.04	3.18	0.90

Notes: The average yearly tradable services (SSIV) employment growth rate is computed following Equation 3.

4 Empirical Strategy

The empirical strategy to estimate multiplier effects is based on the following equation:

$$ln(e_{i,t}^{NTS}) = \beta_0 + \beta_1 ln(e_{i,t}^{TS}) + \mu_i + \eta_t + \varepsilon_{i,t} \tag{2}$$

This baseline estimation captures job multiplier effects in the short term. Given the general equilibrium setting, the specification aims to quantify the effects in a broad, reduced-form approach. That is, to explain levels of non-tradable services (NTS) employment with levels of tradable services (TS). The outcome variable $ln(e_{i,t}^{NTS})$ reflects the log NTS employment in microregion i at time t, and $ln(e_{i,t}^{TS})$ represents the equivalent variable for TS. Additionally, the equation contains microregion fixed effects (μ_i), accounting for unobserved constant local characteristics of these local labour markets, like consumer preferences; and year fixed effects (η_t) to control for time-varying nation-wide shocks in the economy. Therefore, our identification comes from within-region changes in employment, where the coefficient of interest (β_1) reports an elasticity. The job multiplier is derived by multiplying this elasticity by the average NTS-to-TS employment ratio across all periods.

This specification draws largely from Osman and Kemeny (2022). As in their identification strategy, this study relies on frequent data to follow employment levels across time, instead of long-difference specifications. This allows for a more robust implementation of the specification by enabling a better distinction between signal and noise in the coefficient of interest, which becomes less sensitive to arbitrary choices or constraints dictating the first and last years of the analysis. A longer panel also enables the inclusion of robust microregion fixed effects, which is typically not included in specifications partly due to the issue of shorter panels (Bartik and Sotherland, 2019).

Endogeneity concerns can arise in the OLS estimates. Firstly, unobserved evolving microregion characteristics can be simultaneously affecting both NTS and TS employment, such as shifting crime rates or the effects of education policies. Crucially, tradable services are not always traded: a considerable share of employment in TS is driven by local demand. For instance, while bankers can meet non-local demand, this does not mean they exclusively do so; many jobs in banking cater to city residents. These concerns motivate the need to isolate variation reflecting exogenous, non-locally driven TS employment growth, as introduced in table 2.

Following most of the job multiplier literature, this analysis relies on a shift-share instrumental variable (SSIV) to do so. Its classic form takes stock of nation-wide shifts (second half of the equation below, reporting a growth rate) in subsectors (2-digit) of tradable services employment, thus exogenous to specific local demand, and attributes them to microregions based on a weight, or share (the first half of the equation below), of how large those subsectors were - relative to overall TS employment - prior to the shift in a given microregion (see Figure 1).

$$SSIV_{i,t}^{TS} = \sum_{j \in TS}^{J} \frac{e_{i,j,t-1}}{e_{i,J,t-1}} * \left\{ \frac{(E_{j,t} - e_{i,j,t}) - (E_{j,t-1} - e_{i,j,t-1})}{(E_{j,t-1} - e_{i,j,t-1})} \right\} \tag{3}$$

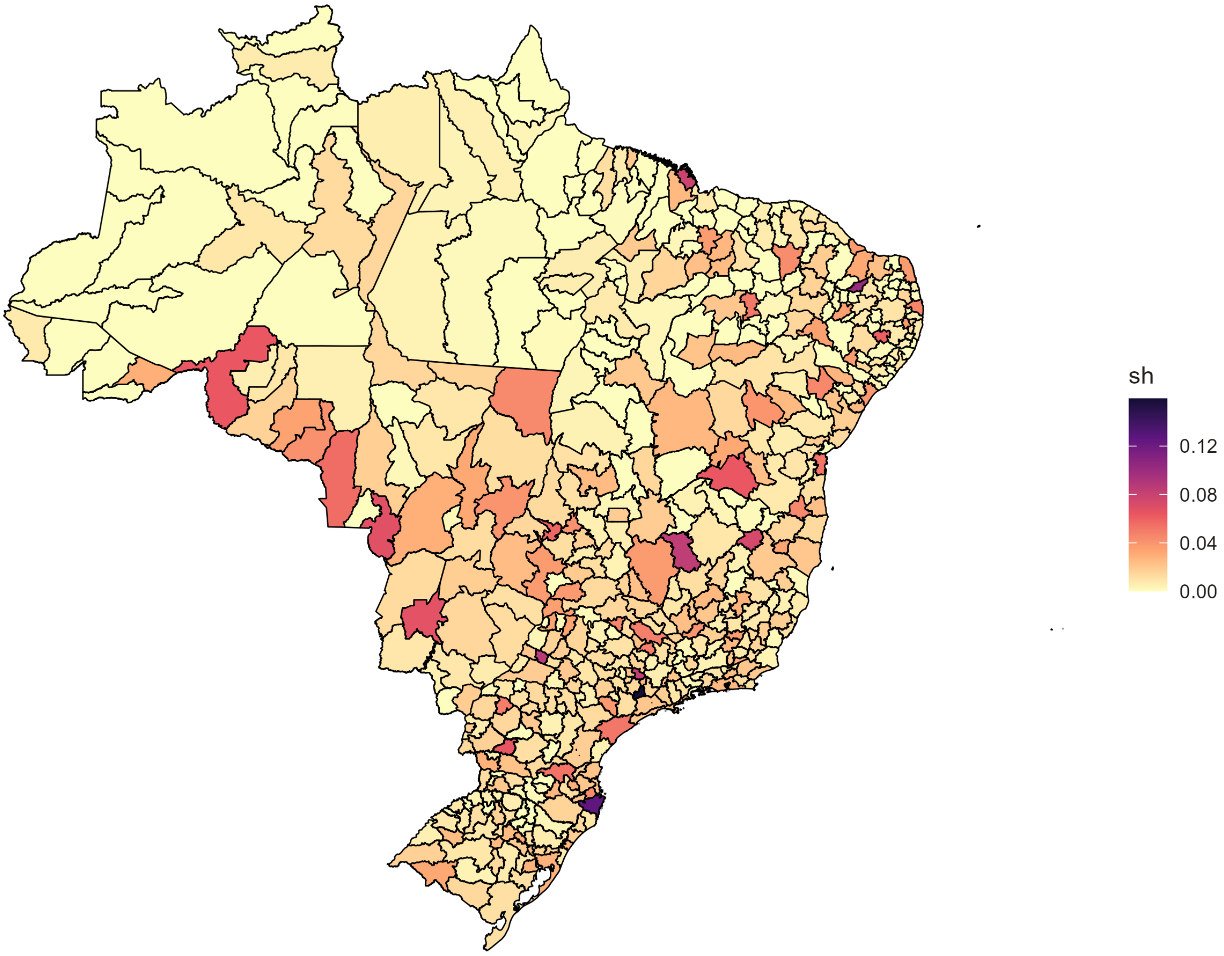
Where j is a 2-digit subsector of tradable services J; $t-1$ and t report the previous and current years of the analysis for a given microregion i. Following Borusyak et al. (2025), a leave-one-out approach is adopted. That is, for a given microregion, the nation-wide shock is computed from the change in national employment in sector j, net of the change in that microregion's own employment in the same sector. This ensures that microregions with a large weight of employment in a given sector cannot influence their allocated exogenous, non-local shift, and is particularly important for sectors that are geographically concentrated, such as tradable services.

Comparing the raw and instrumented series is informative in this respect. While TS employment grew at an average annual rate of 6% across microregions, the corresponding rate implied by the instrument is 3%, which reflects the portion of TS employment growth attributable to non-local rather than local demand (Table 2). The strength of the quasi-national shift as a predictor of a microregion's TS employment growth is itself consistent with these services being tradable, in the sense that competition, prices and demand operate to some degree at a national and international scale. In other words, misclassifying NTS as TS would severely weaken the instrument. If NTS are by definition responding to mostly independent demand trends of each local labour market, the aggregate shift of NTS employment growth would hardly inform on a given microregion's own NTS employment growth.

A key choice in this SSIV specification concerns the definition of pre-exposure shares. While many studies in the job multiplier initially relied on a single long-difference design, now most studies work with stacked differences or, in our case, panel specifications. In settings with multiple periods, Borusyak et al. (2025) note that the exposure shares in the SSIV can either be fixed at the base period or lagged, so as to account for re-specialisation. Given the long span of our panel (1995-2023) and the substantial structural transformation of the services sector in Brazil over this period, we opt to lag shares by one year, in line with Osman and Kemeny (2022). The underlying intuition is that the composition of tradable services has significantly shifted between 1995 and 2023, away from financial services, real estate and land transport and into administrative office, employment and computer programming activities.[12] Therefore, relying on base period (1995) shares to define a microregion's sectoral composition would likely be misrepresentative of its exposure to TS growth in later years due to such structural changes. At the same time, using time-varying shares raises the possibility of endogenous re-specialisation, whereby local shocks may influence sectoral composition. Section 5.4 tests for such threats to the specification with alternative constructions of the instrument.

[12] Table A.2 reports the changes in the 2-digit sector composition of tradable services in Brazil between 1995 and 2023.

Figure 1: Illustrating shares in the SSIV: The weight of IT services in Brazilian microregions, 2008



Source: Relações Anuais de Informações Sociais, Ministry of Labour.

5 Results

5.1 Job multipliers: the nexus between job creation in non-tradable services and in tradable services

Following the identification strategy previously described, we now present estimates for our main variables of interest. Table 3 reports how changes in employment in the tradable sector relate to changes in employment in the non-tradable sector. Since this study aims to focus at a subset of the tradable sector (tradable services), results are compared to effects derived from the overall tradable sector (including manufacturing). Such approach benchmarks our findings to the recent job multiplier literature.

Models 1 and 2 report the OLS results, showing a significant and strong correlation between the rise of jobs in the tradable sector, both including (T) and excluding manufacturing (TS), and jobs in non-tradable services (NTS). Models 1 (and 3) suggest that a 10% increase in TS (T) jobs is associated with a 2% (2.4%) increase in NTS jobs. In terms of job multipliers, since TS is smaller than T by definition, the effect is stronger for TS. For every one new additional job in TS (T), the model predicts 0.9 (0.5) new additional jobs in non-tradable services.

Instrumenting TS employment with the SSIV raises the estimated effects throughout, relative to their OLS counterparts. Strong instruments reinforce the idea that TS are correctly identified, also reflecting the advantage of high-frequency data.[13] In this preferred specification, we also find that the job multiplier of TS is stronger (1.3) than that of T (0.7). These results suggest that strong spillovers of tradable services employment growth on non-tradable services employment should be expected, in particular as employment in tradable services further grows through increased services tradability and services trade liberalization (WTO, 2019).

Table 3: Estimate of the impact of employment in the tradable sector on employment in the non-tradable sector in Brazil

	emp.NTS			
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV
emp.TS	0.202***	0.294**		
	(0.031)	(0.120)		
emp.T			0.240***	0.354*
			(0.041)	(0.196)
F-stat	–	314.49	–	79.64
Adj. R^2	0.967	0.966	0.962	0.940
Observations	14790	14790	14790	14790
FEs	geo, yr	geo, yr	geo, yr	geo, yr

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Because the job multiplier is defined within a general equilibrium setting, the baseline specification deliberately remains reduced-form. However, alternative specifications with controls tend to reinforce that job multiplier effects are strong and significant. Table A.4 reports results with a rich set of base period observables, similar to the exercise of Van Dijk (2018): log employment and population, as well as share of informal workers[14], female workers, workers with college education and workers in manufacturing. Column 4 reports SSIV results with no microregion fixed

[13] A common challenge in the literature, partly driven by low-frequency data, is identifying strong shift-share instrumental variables. Multiple studies report difficulties in dealing with first-stage F-statistics around or below 10 (Moretti, 2010; Van Dijk, 2018)

[14] Since both population and size of the informal market are based on Brazilian Census data, the base period must be set at 1991 instead of 1995.

effects, and column 8 reports SSIV results with state-level (geographic aggregates) fixed effects. While controls are highly significant and help explain where NTS job growth happened in Brazil, they do not weaken the causal link of the job multiplier. Most importantly, results in columns 4 and 8 suggest a similar magnitude of the effect: 1.4 and 1.3 new NTS jobs for every new TS job, respectively.

A further specification replaces base period controls with time-varying ones. Table A.5 compares the baseline results of table 3 with models 3-4, reporting additional controls. These models further show a stable coefficient for the job multiplier. NTS job growth is positively correlated with the share of female and college-educated workers and negatively correlated with the share of manufacturing workers. However, including these covariates does not affect the magnitude of the main covariate, which still points to a 1.3 job multiplier between TS and NTS jobs.

Taken together, these results help characterize the gap between the OLS and SSIV estimates of the job multiplier. In the case of the baseline specification presented in table 3, SSIV coefficients are consistently larger than their OLS counterparts. As previously highlighted, a key feature of tradable services is that they still significantly cater to local demand, which is captured in the OLS estimates. This can both raise an attenuation or upward bias: for instance, in the case of weaker OLS coefficients in the baseline specification, this may signal that TS employment is substantially reliant on local demand and is being partly depressed by microregions with weak local demand. On the other hand, equivalent specifications without microregion fixed effects reported in table A.4 (columns 1-2 and 5-6) point to an upward bias. This may be reflecting how we allow for more co-movement of TS and NTS employment through time-invariant microregion characteristics, such as institutional settings that foster employment. This, in turn, leads to an overestimation of job multiplier effects.

5.2 Mechanisms behind the job multiplier effect in Brazil

Job multipliers can be driven by different cohorts and mechanisms. To compare heterogeneous effects of the job multipliers, we focus on the reported elasticities.

We first decompose TS employment by skill level, based on whether workers hold a college degree. Moretti (2010) and Van Dijk (2018) suggest that growth in high-skilled tradable employment should be the main driver of job multiplier effects, given its association with higher wages. Table

A.6 does not support a clear ranking of this kind in the Brazilian case: the estimated elasticities for high- and low-skilled tradable services employment are of similar magnitude and cannot be statistically distinguished from one another. The interpretation of this result is limited, since the two components of TS employment are closely correlated and are instrumented separately. However, it suggests that spillovers in Brazil are not confined to the expansion of high-skilled tradable employment, and that growth in less skilled segments of the sector also contributes to local job creation.

A follow-up exercise is to understand if households are really the main drivers of the job multiplier, since intersectoral firm-to-firm linkages could also create such spillovers. Based on the OECD Inter Country Input Output tables, we categorise NTS sectors based on whether they cater mainly to household final demand (B2C, business to customer) or firm intermediate consumption (B2B, business to business).[15] In Table A.7, the decomposition suggests that the association between TS employment growth and B2B-intensive NTS employment growth is largely endogenous, as the coefficient loses significance once the instrument is applied. The evidence therefore points to the consumption channel as the leading mechanism behind the job multiplier in Brazil.

Decomposing the remaining non-tradable services helps further identify where job creation takes place in Brazil. Table A.8 reports estimates for construction, retail and government activities, which are not included in either the B2B and B2C-intensive groups. Construction responds most strongly, with an elasticity that roughly doubles once the instrument is applied, while retail — the largest single contributor to non-tradable services employment — also shows a substantial response. Government activities, by contrast, do not respond significantly to tradable services employment growth. This last result is informative for the interpretation of the multiplier: non-market services are largely determined by fiscal and political processes rather than by local household income, so the employment response to an increase in local demand should not operate as flexibly.

[15]This exercise is done through observing how much of a sector's output is sold to either intermediate consumption (other firms) or household final demand. If a sector sells more than half of its output to the former, it is considered B2B-intensive; otherwise, it is considered B2C-intensive. An important nuance is that, for the purpose of these estimates, total output excludes gross fixed capital formation (GFCF), which in the case of services can reflect intangible assets such as software or intellectual property. This is motivated by the fact that GFCF is typically reported as a component of final demand in input-output modelling, which can lead to strong underestimation of firm-to-firm sales. Final categories by sector are reported in Table A.3

5.3 Who benefits most from tradable services job multipliers in Brazil?

Taking stock of the baseline results of the job multiplier in Brazil, we decompose NTS employment to shed light on who wins most from these spillover effects. Distributional effects are to be expected, since the sectoral composition of employment and the consumption patterns of Brazilian households shape which workers benefit from local demand expansions.

Tables A.9 and A.10 report results for a sex-based (men and women) and age-based (under and over 45 years old) decomposition of NTS employment. Our SSIV models indicate that job creation from TS employment growth is concentrated around young men in Brazil. The stronger effects for young men may be linked to a potentially greater responsiveness to local labour demand. One possible explanation is that younger workers face lower mobility or adjustment costs, making them more likely to benefit from local employment expansions. Another possibility is higher relative representation in sectors benefitting more from increased local demand. For instance, the Brazilian construction sector can be particularly prominent as cities face urban expansion, strongly benefitting from increased household and business demand. Demand for particular NTS sectors, some of which have higher relative participation of male workers, may therefore also be a factor at play.

5.4 Alternative outcomes and robustness checks

As is theoretically understood, the job multiplier captures only a partial reaction of the labour market to an exogenous rise in TS employment. Labour mobility is a key determinant to whether additional household income will, for instance, increase employment in NTS. We therefore estimate the impact of TS employment growth on other components of the total wage bill, namely average hours worked and real hourly wage in NTS. We find evidence that both are also positively impacted, in particular real wages: for a 10% increase in TS employment, our model suggests a 2.3% increase in average real wages in NTS (Table A.12).

A further concern is that the estimated multiplier may be driven by a small number of atypical microregions. Brazilian employment is heavily concentrated, and the largest local labour markets could plausibly dominate the results. Re-estimating the baseline after excluding the top and bottom deciles of microregions by employment size leaves the multiplier broadly unchanged (Table A.11), suggesting that the multiplier is not confined to a handful of large metropolitan areas.

As previously highlighted, RAIS lacks information on informal workers, which raises issues questions about whether the employment growth we identify in non-tradable services reflects genuine job creation. To address this, we draw on Brazilian census data to observe total employment, formal and informal, in 2000 and 2010. Following a long-difference specification similar to that of Moretti (2010), we estimate the effect of TS employment growth on non-tradable services employment and on microregion unemployment. The results in Table A.13 suggest that growth in TS employment is associated with higher non-tradable services employment and with a decline in unemployment, which is consistent with the baseline capturing job creation rather than the pure reallocation of informal workers into formal contracts. The estimated elasticity is smaller than in the annual panel, and the instrument is weaker in this specification, which reflects the limitations on an exercise relying on a single long difference across two census cross-sections. Both results nonetheless point in the same direction as the baseline.

Another threat to our specification could be an endogenous SSIV due to lagged shares. As explained, we lag shares by one year, rather than fixing them at the base period, to better account for structural change in tradable services across a long panel. Time-varying shares nonetheless raise the possibility of endogenous re-specialisation. To address this, we re-estimate the instrument using longer lags, so that exposure is defined on increasingly predetermined sectoral composition A.14). The estimated coefficients remain in a range comparable to the baseline, which is difficult to reconcile with endogeneity driven by contemporaneous local shocks. The point estimates also increase modestly with lag length, which may reflect job multiplier effects that materialise over a longer horizon, as more time is allowed for adjustment processes to unfold.

Finally, we try to emulate our results following a similar identification strategy to that of Avdiu et al. (2022). As we have argued, by relying on an exogenous growth of TS jobs, we want to establish a link between services trade and non-tradable job growth in local labour markets. A key advantage of this approach is the detailed sectoral information on services employment, compared to much scarcer information from services trade statistics. As Bartik and Sotherland (2019) show, the robustness of the shift-share instrument improves with the number of underlying sectors, as a more granular decomposition increases the effective number of shocks and reduces the influence of any single sector-specific disturbance. In this sense, our use of detailed service industries helps approximate a "many shocks" environment, strengthening the quasi-experimental variation exploited by the SSIV. Still, the model excludes trade statistics and considers intra-national, cross-microregion flows as trade.

In this exercise, at the expense of lower sectoral detail, we exploit the exogenous variation of services exports as the "shift" component of our SSIV. The growth rate of services exports is obtained from the OECD ICIO (Inter-Country Input-Output) tables published in 2025, which contained services trade information for 21 unique services sectors from 1995 to 2022.[16] To proxy for the rise in Brazilian services exports, we follow Autor et al. ((2020)) and take the growth rate of services exports for a similar basket of Latin American economies.[17] Table A.15 compares baseline results to the ones obtained through a trade-based shift-share instrument. Column 4 suggests a significant and stronger result for the job multiplier, which nearly doubles in size. While magnitudes differ, which is to be expected given the different design with a significantly more aggregate set of industries, results further consolidate the idea that there is a nexus between services trade and job growth in the non-tradable sector. In sum, as services exports increase and generates job growth in tradable services, positive spillover effects are felt across local labour markets.

6 Conclusion

This paper examines whether tradable services generate local employment spillovers and how these compare to those traditionally attributed to the entire tradable sector. Combining Brazilian administrative data with a panel shift-share instrumental variable strategy, we find that employment growth in tradable services may drive substantial job creation in the non-tradable sector.

Identifying tradable services based on their geographic concentration, we estimate a local employment multiplier of approximately 1.3, which exceeds the corresponding multiplier for the broader tradable sector. This figure sits at the lower end of the range reported in earlier work on Brazil and on par with most recent work on advanced economies. More broadly, the results highlight the growing importance of service-based activities that are capable of serving non-local demand in shaping regional labour market dynamics.

We further investigate the mechanisms underlying these effects. The evidence points to the consumption channel as the dominant driver: expansions in tradable services employment increase local household demand for non-tradable goods and services. In contrast, input-output linkages

[16] OECD ICIO services sectors are reported at the ISIC Rev. 4 2-digit level or, often, at more aggregate levels. For services, such sectors are: 41-43; 45-47; 49; 50; 51; 52; 53; 55-56; 58-60; 61; 62-63; 64-66; 68; 69-75; 77-82; 84; 85; 86-88; 90-93; 94-96; and 97-98. The full list of ISIC Rev. 4 2-digit sectors is reported in table A.1

[17] This is obtained through a weighted average, based on exports, of the following countries available in the OECD ICIO 2025: Argentina, Mexico, Colombia, Chile, Costa Rica and Peru.

between firms appear to play a more limited role in the job multiplier. Our findings also question the conventional expectation that high-skilled tradable employment is the principal source of job multipliers: we cannot statistically distinguish the elasticities associated with high- and low-skilled tradable services employment, which suggests that spillovers are not confined to the expansion of skill-intensive segments of the sector.

The gains from tradable services employment growth are not evenly distributed. Spillovers are stronger for male workers and for younger workers, reflecting the sectoral composition of job creation in Brazil. Complementary results on wages and hours worked indicate that local labour markets adjust along multiple margins, reinforcing the overall importance of these spillovers.

These findings have implications for regional and labour market policy. As economies, both advanced and emerging ones such as Brazil, continue to shift towards service-based production, understanding the spillover effects from sectors exposed to non-local demand becomes increasingly important. While policy discussions have traditionally focused on manufacturing, our results suggest that tradable services can also act as significant drivers of local employment growth.

More broadly, although our baseline empirical design does not directly observe trade flows, the results are consistent with the increasing tradability of services — including through cross-border trade — having meaningful local labour market implications. This matters for how spatial disparities are understood. Tradable services are considerably more concentrated across the Brazilian territory than non-tradable services, so the local gains from their expansion accrue unevenly across regions, and the spillovers documented here operate on top of an already uneven distribution of activity. Whether that pattern persists is an open question, since the same technologies raising the tradability of services are also reshaping where the workers producing them live and spend their income. The diffusion of remote work is the clearest instance: where tradable services workers no longer commute to the places they are employed, the local demand sustaining non-tradable employment may be redistributed away from those places altogether (Althoff et al., 2022). How the spillovers documented here evolve under such shifts, and whether they continue to reinforce or begin to attenuate existing spatial disparities, is a question that the growing availability of high-frequency administrative data should make it possible to address.

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A Services categories

Table A.1: Categorising services tradable and non-tradable

Section (A-U)	Division (2-digit)	(1)	(2)	(3)	Category
Construction (F)	41. Construction of buildings	0.22	No	No	Non-tradable
	42. Civil engineering	0.3	No	No	Non-tradable
	43. Specialized construction activities	0.63	Yes	No	Non-tradable
Wholesale and retail (G)	45. Wholesale and retail trade and repair of motor vehicles and motorcycles	0.19	No	Yes	Non-tradable
	46. Wholesale trade, except of motor vehicles and motorcycles	0.11	No	Yes	Non-tradable
	47. Retail trade, except of motor vehicles and motorcycles	0.03	No	Yes	Non-tradable
Transportation and storage (H)	49. Land transport and transport via pipelines	0.12	No	No	Tradable
	50. Water transport	11.49	Yes	Yes	Tradable
	51. Air transport	6.73	Yes	Yes	Tradable
	52. Warehousing and support activities for transportation	0.78	Yes	Yes	Tradable
	53. Postal and courier activities	0.23	No	Yes	Non-tradable
Accommodation and food services (I)	55. Accommodation	0.9	Yes	No	Non-tradable
	56. Food and beverage service activities	0.3	No	No	Non-tradable
Information and communication (J)	58. Publishing activities	3.51	Yes	Yes	Tradable
	59. Motion picture, video and television programme production, sound recording and music publishing activities	3.42	Yes	No	Tradable
	60. Programming and broadcasting activities	0.89	Yes	Yes	Tradable
	61. Telecommunications	0.22	No	No	Tradable
	62. Computer programming, consultancy and related activities	3.1	Yes	Yes	Tradable
	63. Information service activities	3.76	Yes	Yes	Tradable
Finance and insurance (K)	64. Financial service activities, except insurance and pension funding	2.05	Yes	No	Tradable
	65. Insurance, reinsurance and pension funding, except compulsory social security	4.9	Yes	Yes	Tradable
	66. Activities auxiliary to financial service and insurance activities	9.48	Yes	Yes	Tradable
Real estate (L)	68. Real estate activities	0.25	No	No	Tradable
Professional, scientific and technical activities (M)	69. Legal and accounting activities	0.07	No	No	Tradable
	70. Activities of head offices; management consultancy activities	6.62	Yes	Yes	Tradable
	71. Architectural and engineering activities; technical testing and analysis	0.61	Yes	Yes	Tradable
	72. Scientific research and development	2.72	Yes	Yes	Tradable

Continued on next page

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Section (A-U)	Division (2-digit)	(1)	(2)	(3)	Category
	73. Advertising and market research	3.26	Yes	Yes	Tradable
	74. Other professional, scientific and technical activities	4.08	Yes	No	Tradable
	75. Veterinary activities	0.22	No	No	Non-tradable
Administrative and support services (N)	77. Rental and leasing activities	0.79	Yes	No	Non-tradable
	78. Employment activities	4.01	Yes	Yes	Tradable
	79. Travel agency, tour operator, reservation service and related activities	1.79	Yes	Yes	Tradable
	80. Security and investigation activities	1.3	Yes	No	Non-tradable
	81. Services to buildings and landscape activities	1.71	Yes	No	Non-tradable
	82. Office administrative, office support and other business support activities	1.07	Yes	Yes	Tradable
Public administration and defence (O)	84. Public administration and defence; compulsory social security	0.96	Yes	No	Non-tradable
Education (P)	85. Education	0.2	No	No	Non-tradable
Human health and social work activities (Q)	86. Human health activities	0.18	No	Yes	Non-tradable
	87. Residential care activities	0.39	No	No	Non-tradable
	88. Social work activities without accommodation	2.35	Yes	No	Non-tradable
Arts, entertainment and recreation (R)	90. Creative, arts and entertainment activities	1.53	Yes	No	Non-tradable
	91. Libraries, archives, museums and other cultural activities	2.54	Yes	No	Non-tradable
	92. Gambling and betting activities	2.91	Yes	No	Non-tradable
	93. Sports activities, amusement and recreation activities	0.28	No	No	Non-tradable
Other service activities (S)	94. Activities of membership organizations	0.39	No	Yes	Non-tradable
	95. Repair of computers and personal and household goods	0.55	Yes	No	Non-tradable
	96. Other personal service activities	0.15	No	No	Non-tradable
Households activities (T)	97. Activities of households as employers of domestic personnel	3.42	Yes	No	Non-tradable
	98. Undifferentiated goods- and services-producing activities of private households for own use	Not included in RAIS			Non-tradable
Extraterritorial bodies (U)	99. Activities of extraterritorial organizations and bodies	43.3	Yes	No	Non-tradable

Notes: Column (1) reports the geographic concentration of the sector in 2023, following equation 1. Column (2) indicates whether the sector is above the tradability cutoff (lowest geographic concentration in manufacturing). This cutoff in 2023 refers to division 33 (Repair and installation of machinery and equipment), which reported a geographic concentration of 0.44. Column (3) indicates whether the sector is consistently above or below the cutoff for every year between 1994 and 2023. Divisions whose classification is not constant across all years of the sample are assigned manually on the basis of their ISIC Rev. 4 descriptions, which accounts for the four cases reported as tradable despite a 2023 concentration value below the cutoff: land transport (49), telecommunications (61), real estate (68), and legal and accounting activities (69). In each case the activity is capable of supplying clients outside the microregion where it is produced, and its relatively even spatial distribution likely reflects the geography of demand rather than an inability to serve that demand at distance.

Table A.2: Composition of tradable services in Brazil by shares of 2-digit sector employment, 1995 and 2023

Section (A-U)	Division (2-digit)	1995	2023	Difference
Transportation and storage (H)	49. Land transport and transport via pipelines	27.5%	20.2%	-7.3
	50. Water transport	1.0%	0.5%	-0.5
	51. Air transport	1.3%	0.6%	-0.7
	52. Warehousing and support activities for transportation	2.5%	5.3%	2.8
Information and communication (J)	58. Publishing activities	1.7%	2.4%	0.7
	59. Motion picture, video and television programme production, sound recording and music publishing activities	2.8%	0.3%	-2.5
	60. Programming and broadcasting activities	1.0%	0.8%	-0.2
	61. Telecommunications	3.5%	2.9%	-0.6
	62. Computer programming, consultancy and related activities	0.9%	4.7%	3.8
	63. Information service activities	1.8%	1.4%	-0.4
Finance and insurance (K)	64. Financial service activities, except insurance and pension funding	17.7%	7.3%	-10.4
	65. Insurance, reinsurance and pension funding, except compulsory social security	1.7%	2.0%	0.3
	66. Activities auxiliary to financial service and insurance activities	1.3%	1.9%	0.6
Real estate (L)	68. Real estate activities	9.8%	2.1%	-7.7
Professional, scientific and technical activities (M)	69. Legal and accounting activities	4.0%	5.6%	1.6
	70. Activities of head offices; management consultancy activities	3.2%	1.9%	-1.3
	71. Architectural and engineering activities; technical testing and analysis	1.3%	4.3%	3.0
	72. Scientific research and development	1.3%	0.6%	-0.7
	73. Advertising and market research	0.9%	1.9%	1.0
	74. Other professional, scientific and technical activities	0.4%	1.7%	1.3
Administrative and support services (N)	78. Employment activities	4.0%	9.4%	5.4
	79. Travel agency, tour operator, reservation service and related activities	3.2%	1.9%	-1.9
	82. Office administrative, office support and other business support activities	9.2%	21.5%	12.3

Notes: Columns 1995 and 2023 report the the given sector (division)'s employment share of tradable services for each respective year. The last column (difference) reports percentage point differences between both years.

Table A.3: Categorising market non-tradable services into business-to-business- and business-to-consumer-intensive

Section (A-U)	Division (2-digit)	(1)	Category
Transportation and storage (H)	53. Postal and courier activities	92.4%	B2B
Accommodation and food services (I)	55. Accommodation 56. Food and beverage service activities	18.3%	B2C
Professional, scientific and technical activities (M)	75. Veterinary activities	-	B2C
Administrative and support services (N)	77. Rental and leasing activities 80. Security and investigation activities 81. Services to buildings and landscape activities	87.7%	B2B
Education (P)	85. Education	0.9%	B2C
Human health and social work activities (Q)	86. Human health activities 87. Residential care activities 88. Social work activities without accommodation	1.7%	B2C
Arts, entertainment and recreation (R)	90. Creative, arts and entertainment activities 91. Libraries, archives, museums and other cultural activities 92. Gambling and betting activities 93. Sports activities, amusement and recreation activities	7.5%	B2C
Other service activities (S)	94. Activities of membership organizations 95. Repair of computers and personal and household goods 96. Other personal service activities	9.7%	B2C
Households activities (T)	97. Activities of households as employers of domestic personnel 98. Undifferentiated goods- and services-producing activities of private households for own use	0%	B2C

Notes: Column (1) reports the share of output (excluding gross fixed capital formation) of the sector that is destined to intermediate consumption of other firms. Results are estimated using the OECD Inter Country Input-Output tables, which contain less sector detail than RAIS. Because results are available only at the section level, and veterinary activities are bundled within a largely tradable, B2B-intensive section, no value is reported for this sector. Given its ISIC Rev. 4 description, it is treated as B2C-intensive.

B Additional results

Table A.4: Alternative specifications: base period controls

	emp.NTS (without geographic FE)				*emp.NTS* (with state FE)			
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV	(5) OLS	(6) SSIV	(7) OLS	(8) SSIV
emp.TS	0.649***	0.529***	0.339***	0.325***	0.713***	0.574***	0.345***	0.309***
	(0.014)	(0.025)	(0.020)	(0.066)	(0.012)	(0.034)	(0.021)	(0.069)
95.*sh.eduhigh*			1.700***	1.750***			1.480***	1.540***
			(0.242)	(0.273)			(0.284)	(0.317)
95.*sh.mfgemp*			-0.541***	-0.554***			-0.535***	-0.559***
			(0.111)	(0.125)			(0.096)	(0.109)
95.*sh.fememp*			0.670***	0.635**			0.417**	0.374*
			(0.123)	(0.198)			(0.146)	(0.168)
95.*logemp*			0.317***	0.329***			0.393***	0.427***
			(0.026)	(0.066)			(0.030)	(0.073)
91.*logpop*			0.222***	0.225***			0.139***	0.147***
			(0.027)	(0.028)			(0.025)	(0.029)
91.*sh.infemp*			0.397**	0.389**			0.325*	0.326*
			(0.137)	(0.132)			(0.139)	(0.145)
F-stat	–	697.49	–	313.36	–	421.34	–	360.13
Adj. R^2	0.866	0.841	0.940	0.940	0.912	0.889	0.950	0.950
Observations	14,790	14,790	13,572	13,572	14,790	14,790	13,572	13,572
FEs	yr	yr	yr	yr	yr, geo (state)	yr, geo (state)	yr, geo (state)	yr, geo (state)

Standard errors clustered at the GEO level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Samples with full set of controls are affected, since not all Brazilians microregions are accurately identifiable in the Brazilian 1991 Census. Variables for population and share of informal employment are taken from this source.

Table A.5: Alternative specifications: time-varying controls

	$emp.NTS$ (baseline)		$emp.NTS$ (with controls)	
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV
$emp.TS$	0.202***	0.294*	0.201***	0.298**
	(0.031)	(0.120)	(0.027)	(0.107)
$sh.eduhigh$			2.17***	2.12***
			(0.276)	(0.283)
$sh.mfgemp$			-0.682***	-0.621**
			(0.180)	(0.194)
$sh.fememp$			1.43***	1.60***
			(0.366)	(0.398)
F-stat	–	314.49	–	205.78
Adj. R^2	0.967	0.966	0.974	0.973
Observations	14790	14790	14790	14790
FEs	geo, yr	geo, yr	geo, yr	geo, yr

Standard errors clustered at the GEO level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table A.6: Heterogeneity: skills

	$emp.NTS$			
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV
High-skilled ($emp.TS$)	0.248***	0.187**		
	(0.017)	(0.083)		
Low-skilled ($emp.TS$)			0.055***	0.202***
			(0.017)	(0.060)
F-stat	–	120.85	–	264.59
Adj. R^2	0.970	0.970	0.965	0.962
Observations	14790	14790	14790	14790
FEs	geo, yr	geo, yr	geo, yr	geo, yr

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.7: Mechanisms: sectoral split of non-tradable services

	B2B-intensive ($emp.NTS$)		B2C-intensive ($emp.NTS$)	
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV
$emp.TS$	0.358***	0.212	0.301***	0.356***
	(0.042)	(0.149)	(0.030)	(0.108)
F-stat	–	314.49	–	314.49
Adj. R^2	0.941	0.938	0.974	0.971
Observations	14790	14790	14790	14790
FEs	geo, yr	geo, yr	geo, yr	geo, yr

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.8: Mechanisms: sectoral split of non-tradable services (2)

	Construction ($emp.NTS$)		Retail ($emp.NTS$)		Government ($emp.NTS$)	
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV	(5) OLS	(6) SSIV
$emp.TS$	0.516***	0.990***	0.317***	0.772***	0.161***	0.259
	(0.054)	(0.242)	(0.029)	(0.137)	(0.055)	(0.179)
F-stat	–	314.49	–	314.49	–	314.49
Adj. R^2	0.879	0.873	0.978	0.975	0.891	0.889
Observations	14790	14790	14790	14790	14790	14790
FEs	geo, yr	geo, yr	geo, yr	geo, yr	geo, yr	geo, yr

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.9: Heterogeneity: sex

	Men ($emp.NTS$)		Women ($emp.NTS$)	
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV
$emp.TS$	0.255***	0.513***	0.160***	0.121
	(0.029)	(0.130)	(0.036)	(0.122)
F-stat	–	314.49	–	314.49
Adj. R^2	0.968	0.964	0.959	0.959
Observations	14790	14790	14790	14790
FEs	geo, yr	geo, yr	geo, yr	geo, yr

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.10: Heterogeneity: age

	Under-45 ($emp.NTS$)		45+ ($emp.NTS$)	
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV
$emp.TS$	0.219***	0.358***	0.174***	0.071
	(0.031)	(0.125)	(0.034)	(0.103)
F-stat	–	314.49	–	314.49
Adj. R^2	0.965	0.964	0.967	0.966
Observations	14790	14790	14790	14790
FEs	geo, yr	geo, yr	geo, yr	geo, yr

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.11: Baseline model, without bottom and top decile of microregions (by employment)

	$emp.NTS$			
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV
$emp.TS$	0.220***	0.249**		
	(0.030)	(0.097)		
$emp.T$			0.147***	0.585***
			(0.042)	(0.196)
F-stat	–	101.86	–	22.83
Adj. R^2	0.940	0.940	0.922	0.886
Observations	9454	9454	7540	7540
FEs	geo, yr	geo, yr	geo, yr	geo, yr

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.12: Beyond employment: additional impacts of growth in employment in tradable services

	Log hours per NTS worker		Log real NTS hourly wage	
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV
$emp.TS$	0.016***	0.054***	0.068***	0.235***
	(0.003)	(0.019)	(0.010)	(0.049)
F-stat	–	148.24	–	148.24
Adj. R^2	0.618	0.570	0.866	0.835
Observations	12240	12240	12240	12240
FEs	geo, yr	geo, yr	geo, yr	geo, yr

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Table A.13: Including the informal sector: the impact of services trade on unemployment (2000 and 2010)

	$\Delta emp.NTS$		Unemployment rate	
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV
$\Delta emp.TS$	0.113**	0.176***	-1.292***	-5.558*
	(0.036)	(0.049)	(0.265)	(2.677)
F-stat	–	11.8	–	11.1
Adj. R^2	0.017	0.005	0.020	0.017
Observations	510	510	510	510
FEs				

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table A.14: Lagged shares

	$emp.NTS$	
	(1) SSIV (2-y lag)	(2) SSIV (3-y lag)
$emp.TS$	0.219*	0.261**
	(0.123)	(0.118)
F-stat	177.19	184.12
Adj. R^2	0.972	0.972
Observations	13770	13770
FEs	geo, yr	geo, yr

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Both models are estimated on the common sample of years for which the full lag structure is available (1997–2023), so that lag lengths are not shortened in early years. The baseline specification, with a one-year lag on the full 1995–2023 sample, is reported in Table **??**.

Table A.15: Alternative instrument: trade-based shock

	$emp.NTS$ (baseline)		$emp.NTS$ (trade-based shock)	
	(1) OLS	(2) SSIV	(3) OLS	(4) SSIV
$emp.TS$	0.202***	0.294**	0.179***	0.521***
	(0.031)	(0.120)	(0.030)	(0.120)
F-stat	–	314.49	–	188.84
Adj. R^2	0.967	0.966	0.970	0.964
Observations	14790	14790	13770	13770
FEs	geo, yr	geo, yr	geo, yr	geo, yr

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. SEs clustered at microregion level.

Columns 3-4 are based on a trade-based IV, which uses data only available from 1995 to 2022. Therefore, the sample is smaller due to the loss of information for 1995 and 2023.